

MIT Academy of Engineering, Alandi(D), Pune
Department of CSE AIML

Mini Project

Course Code: 2311162L

Course Name: Data Visualization Lab

Academic Year: 2025–26

Student Details

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- **Division / Batch:** AL2/A25
- **Dataset Name & Source:** Global Air Pollution Analysis
- **Dataset Source Link:** <https://www.kaggle.com/datasets/sazidthe1/global-air-pollution-data>
- **Video Explanation by Team Members:**
<https://drive.google.com/file/d/1C65440X5gKz81Shdm7L9YkE-yyUKp2wA/view?usp=sharing>
- **Power BI Service Dashboard Link:**
https://app.powerbi.com/links/Fy9zvYaATT?ctid=7e394fc8-4b86-4cfe-810e-43f86f8bec47&pbi_source=linkShare

Air Pollution Analysis Dashboard

1. Introduction

In today's world, air pollution is one of the most critical environmental challenges affecting public health globally. This project focuses on analyzing a global air pollution dataset to uncover meaningful insights related to pollutant levels, AQI categories, and country-wise pollution distribution.

Using Power BI, an interactive multi-page dashboard was created to visualize and analyze the dataset efficiently. The project aims to help understand trends such as AQI distribution, pollutant concentration by country, and risk levels across different regions.

2. Objectives

- I. To analyze global air pollution data using visualization techniques
- II. To identify patterns in AQI levels and pollutant distribution across countries
- III. To create an interactive dashboard for better understanding of air quality
- IV. To apply Data Visualization concepts using Power BI

3. Dataset Description

The dataset used in this project is a global air pollution dataset obtained from Kaggle. It contains the following attributes:

1. City
2. Country
3. AQI Value
4. AQI Category

5. CO AQI Value
6. CO AQI Category
7. NO2 AQI Value
8. NO2 AQI Category
9. Ozone AQI Value
10. Ozone AQI Category
11. PM2.5 AQI Value
12. PM2.5 AQI Category

The screenshot shows the Microsoft Power Query Editor interface. A 'Choose Columns' dialog box is open, allowing the user to select which columns to keep in the resulting table. The dialog lists the following columns: (Select All Columns), country, city, aqi_value, aqi_category, co_aqi_value, co_aqi_category, ozone_aqi_value, ozone_aqi_category, no2_aqi_value, no2_aqi_category, pm25_aqi_value, and pm25_aqi_category. The 'OK' button is highlighted. In the background, the 'air_pollution_data' table is visible, showing columns for country, city, and various AQI values and categories. The table has 114 distinct values for country and 1000 distinct values for city.

AirPollution_Dashboard_204and213 • Last saved: 5/9/2026 at 10:28 PM • SAHIL LOHIANDE

File Home Help Table tools

Name: air_pollution_data

Manage relationships | New measure | Quick measure | New table | Mark as date table | Calculations | Relationships | Calendars

country	city	aqi_value	aqi_category	co_aqi_value	co_aqi_category	ozone_aqi_value	ozone_aqi_category	no2_aqi_value	no2_aqi_category	pm25_aqi_value
Russian Federation	Polevskoy	31	Good	1	Good	31	Good	0	Good	
Russian Federation	Gukovo	38	Good	1	Good	38	Good	0	Good	
Russian Federation	Tyukalinsk	34	Good	1	Good	34	Good	0	Good	
Russian Federation	Glubokiy	35	Good	1	Good	35	Good	0	Good	
Russian Federation	Zmeinogorsk	35	Good	1	Good	35	Good	0	Good	
Russian Federation	Cherepanovo	38	Good	1	Good	38	Good	0	Good	
Russian Federation	Cherkessk	49	Good	1	Good	40	Good	0	Good	
Russian Federation	Rostov	32	Good	1	Good	32	Good	0	Good	
Russian Federation	Desnogorsk	34	Good	1	Good	34	Good	0	Good	
Russian Federation	Donskoye	37	Good	1	Good	37	Good	0	Good	
Russian Federation	Kinel Cherkassy	35	Good	1	Good	35	Good	0	Good	
Russian Federation	Kola	30	Good	1	Good	30	Good	0	Good	
Russian Federation	Komsomolsk Na Amure	42	Good	1	Good	32	Good	0	Good	
Russian Federation	Kotelnikovo	39	Good	1	Good	39	Good	0	Good	
Russian Federation	Krasnovishersk	29	Good	1	Good	28	Good	0	Good	
Russian Federation	Kalyazin	24	Good	1	Good	24	Good	0	Good	
Russian Federation	Nizhniye Sergi	37	Good	1	Good	35	Good	0	Good	
Russian Federation	Kormilovka	38	Good	1	Good	38	Good	0	Good	
Russian Federation	Karabash	36	Good	1	Good	36	Good	0	Good	
Russian Federation	Kotelnich	33	Good	1	Good	33	Good	0	Good	
Russian Federation	Krasnokamsk	25	Good	1	Good	25	Good	0	Good	
Russian Federation	Kurtamysh	31	Good	1	Good	31	Good	0	Good	
Russian Federation	Markha	36	Good	1	Good	36	Good	0	Good	
Russian Federation	Mikhaylovka	43	Good	1	Good	31	Good	0	Good	
Russian Federation	Nikel	32	Good	1	Good	32	Good	0	Good	
Russian Federation	Omega	35	Good	1	Good	35	Good	0	Good	
Russian Federation	Pavlovo	34	Good	1	Good	34	Good	0	Good	

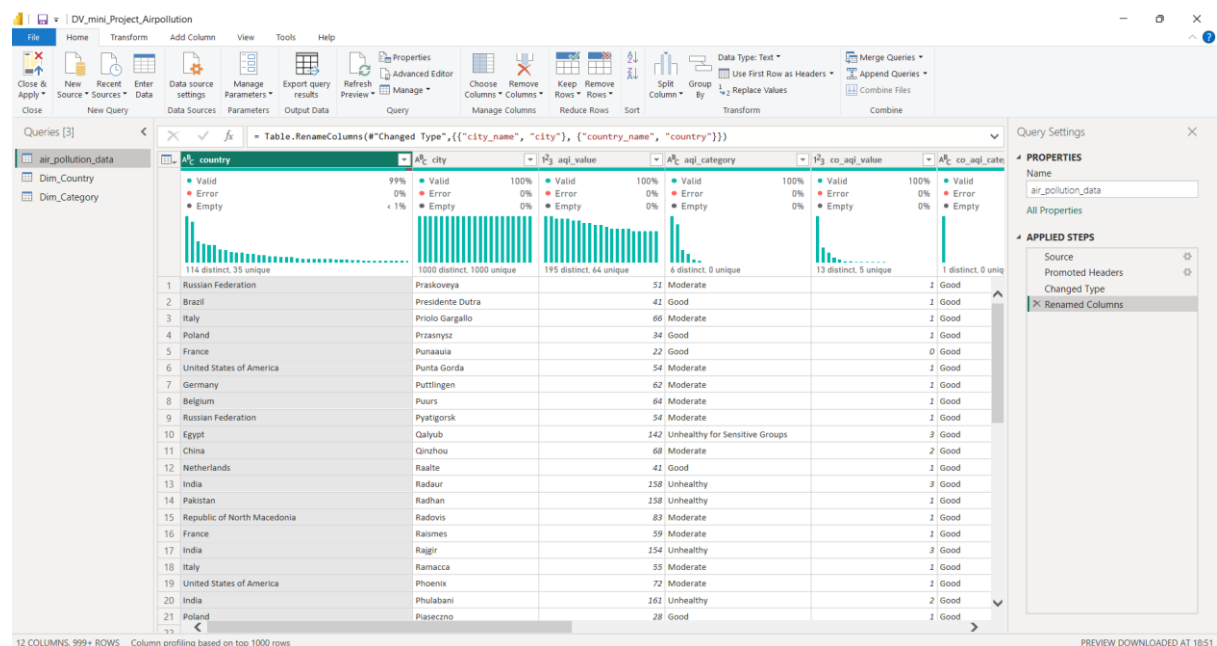
Table: air_pollution_data (23,463 rows)

Data pane on the right shows the table structure and available fields: air_pollution_data, aqi_category, aqi_value, Avg AQI Category, city, co_aqi_category, co_aqi_value, country, Country-City, Date, Max AQI, Max Co, Max NO2, Ozone, Pm2.5, no2_aqi_category, no2_aqi_value, ozone_aqi_category, ozone_aqi_value, pm25_aqi_category, pm25_aqi_value, pm25_aqi_value, Calendar, AQI MTD, AQI QTD, AQI Running Total.

4. Data Cleaning & Preparation

The dataset was cleaned using Power BI Power Query:

- Removed duplicate and blank rows
- Handled missing values in pollutant columns
- Standardized AQI category labels
- Converted data types (AQI Value to Number, Category columns to Text)
- Created calculated columns for risk classification



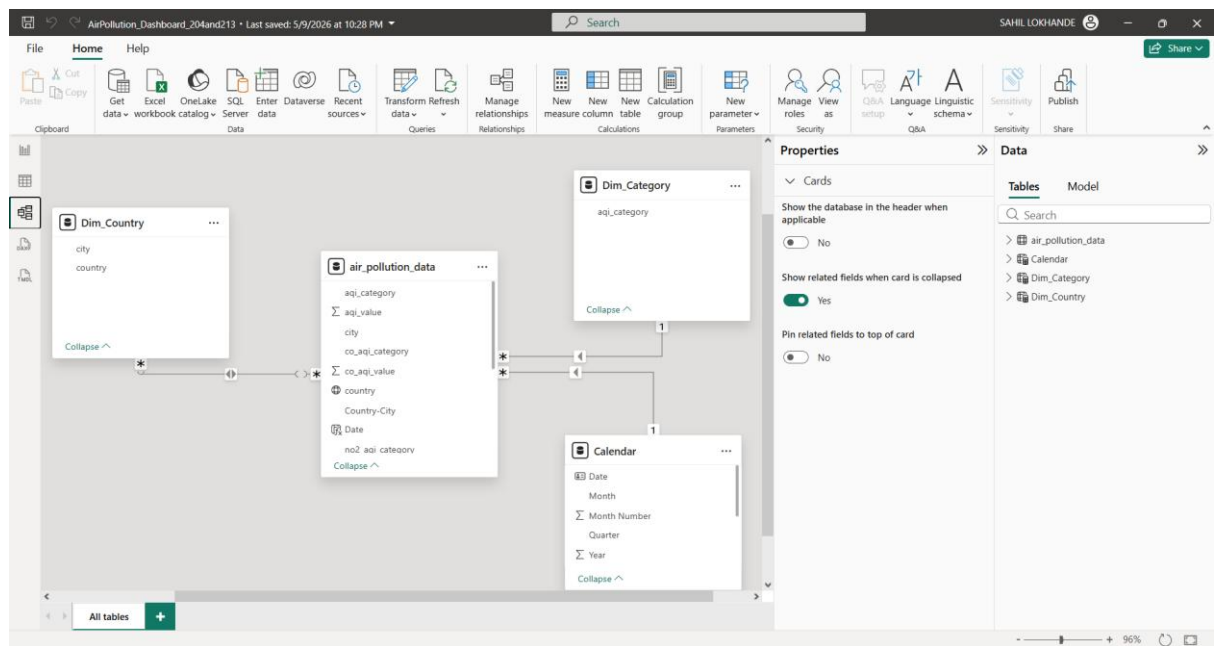
5. Data Modeling

A star schema approach was followed:

Fact Table: air_pollution_data

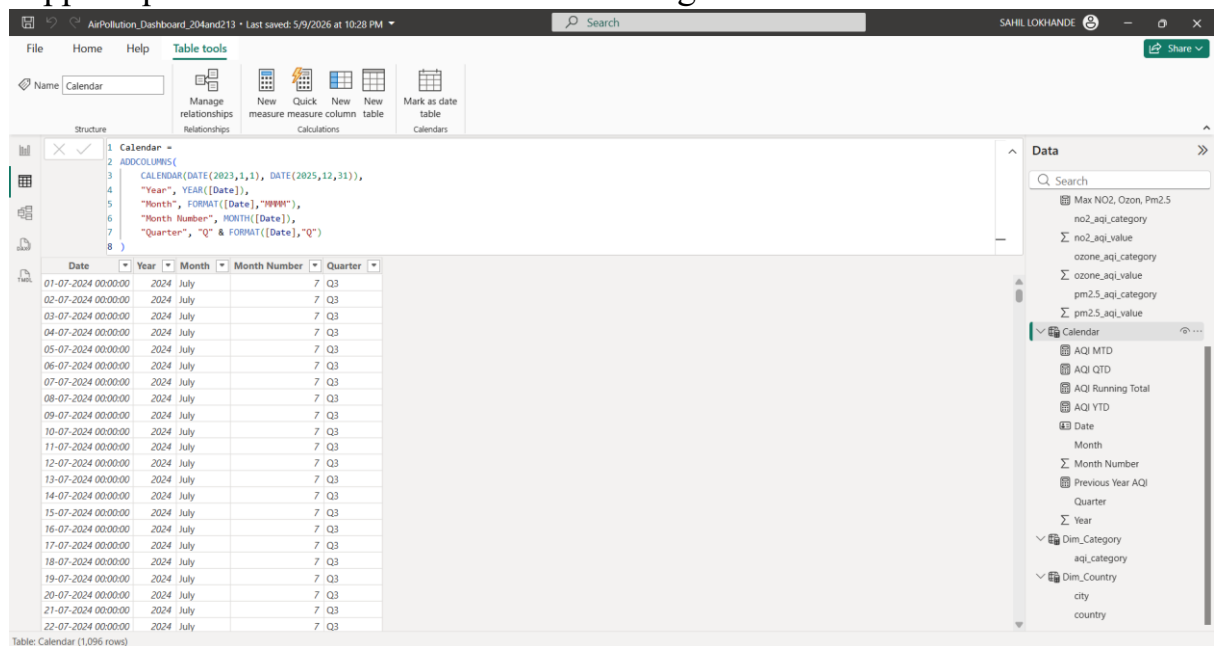
Dimension Tables: Dim_Category, Dim_Country, Calendar Table

Relationships were created between the fact table and dimension tables using city, country, and AQI category fields.



6. Calendar Table

A Calendar table was created using DAX to enable time-based analysis and support quarter-wise and month-wise filtering of data.



7. Measures Created (DAX)

Several measures were created to analyze data:

- Avg AQI
- Max AQI
- Min AQI
- Avg NO2
- Avg CO
- Avg Ozone
- Avg PM2.5
- Hazardous Count
- Overall Pollution Score
- Above Threshold

These measures help summarize and compare pollution levels across countries.

The screenshot displays the Power BI interface with two main panes: Visualizations and Data.

Visualizations Pane:

- Build visual:** Includes a search bar and a grid of visualization icons (bar, line, pie, etc.).
- Values:** A dashed box with the text "Add data fields here".
- Drill through:** A section with a "Cross-report" toggle (set to Off) and a "Keep all filters" toggle (set to On).
- Add drill-through fields here:** A dashed box for adding fields to the drill-through view.

Data Pane:

- air_pollution_data:** A table with various fields, each with a checkbox:
 - ☐ aqi_category
 - ☐ Σ aqi_value
 - ☐ Avg AQI Categ...
 - ☐ city
 - ☐ co_aqi_category
 - ☐ Σ co_aqi_value
 - ☐ country
 - ☐ Country-City
 - ☐ Date
 - ☐ Max AQI
 - ☐ Max Co
 - ☐ Max NO2, Ozo...
 - ☐ no2_aqi_category
 - ☐ Σ no2_aqi_value
 - ☐ ozone_aqi_cate...
 - ☐ Σ ozone_aqi_value
 - ☐ pm2.5_aqi_cate...
 - ☐ Σ pm2.5_aqi_value
- Calendar:** A table with a checkbox and a right arrow.
- Dim_Category:** A table with a checkbox and a right arrow.
- Dim_Country:** A table with a checkbox and a right arrow.

8. Calculated Columns

New columns were created to enhance analysis:

AQI Risk Level (Good, Moderate, Unhealthy, Very Unhealthy, Hazardous)

CO Risk (High/Normal)

NO2 Risk (High/Normal)

Ozone Risk (High/Normal)

Above Threshold flag

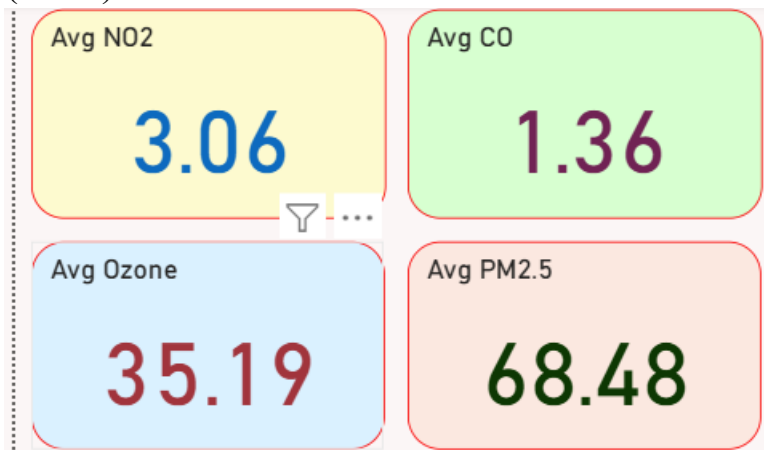
These columns help categorize and segment air quality data by risk.

9. Data Visualization (Dashboard)

The dashboard includes multiple visuals across pages:

KPI Cards

Display Avg NO2 (3.06), Avg CO (1.37), Avg Ozone (35.19), Avg PM2.5 (68.52)

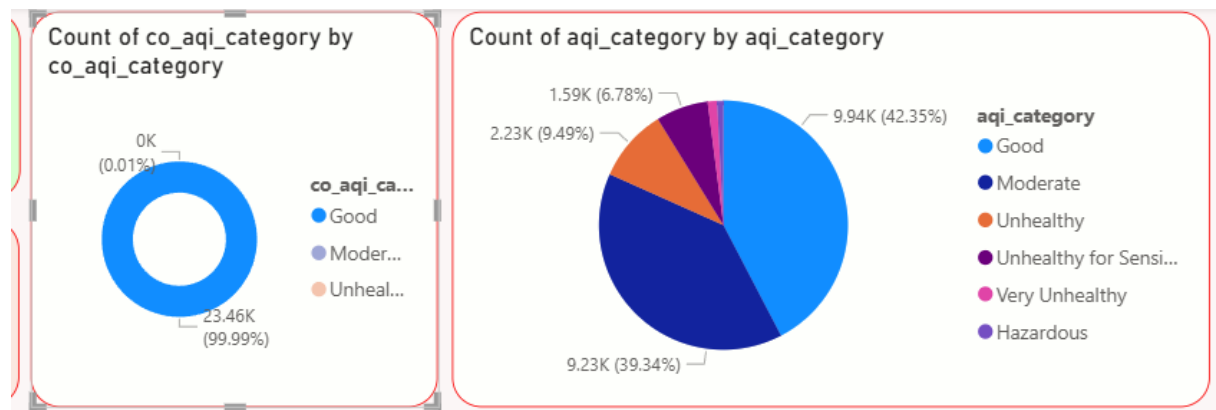


Pie / Donut Chart

Shows distribution of AQI categories (Good, Moderate, Unhealthy, Hazardous, etc.)

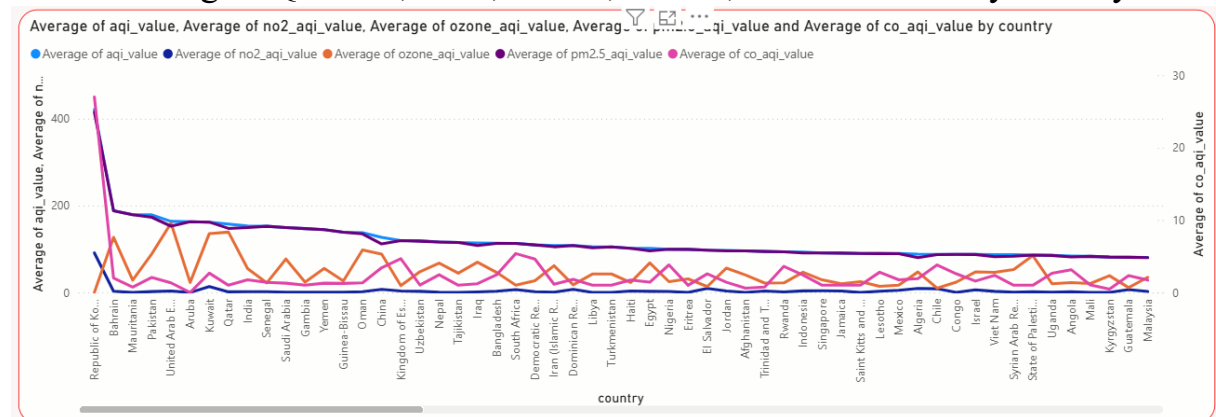
Shows pollutant contribution breakdown (PM2.5 dominates at 38.03%, AQI avg at 39.97%)

Displays CO AQI category distribution



Line Chart

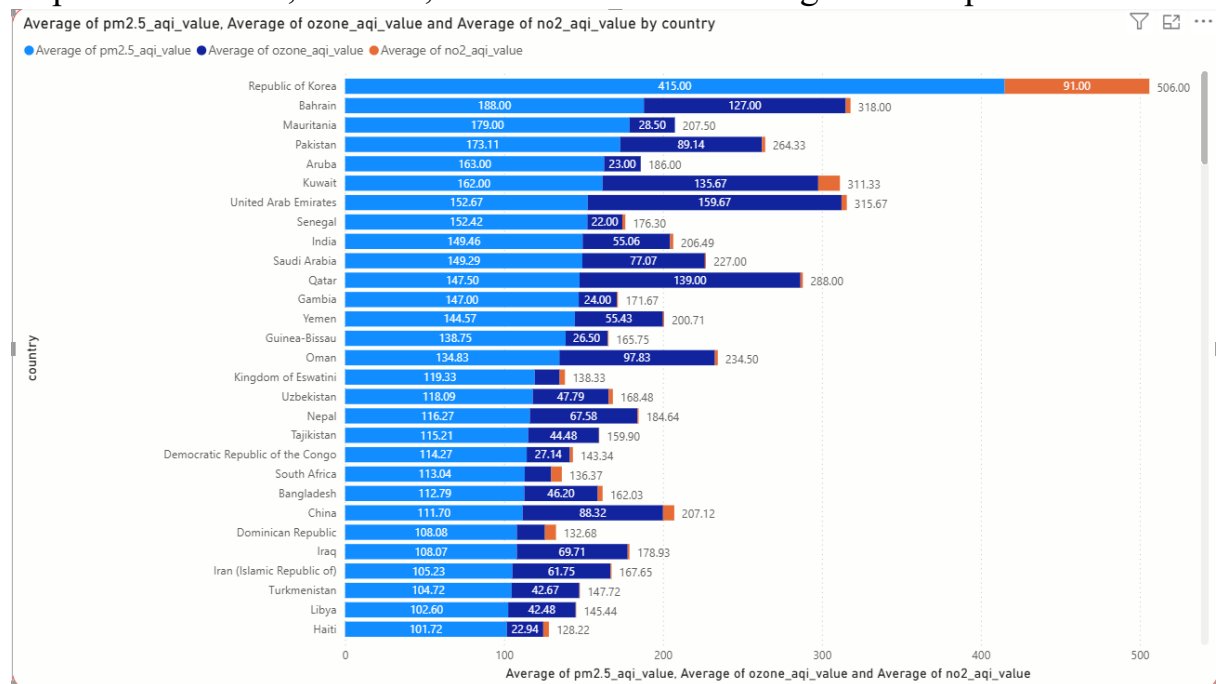
Shows average AQI value, NO2, Ozone, PM2.5, and CO trends by country



Clustered Bar Chart

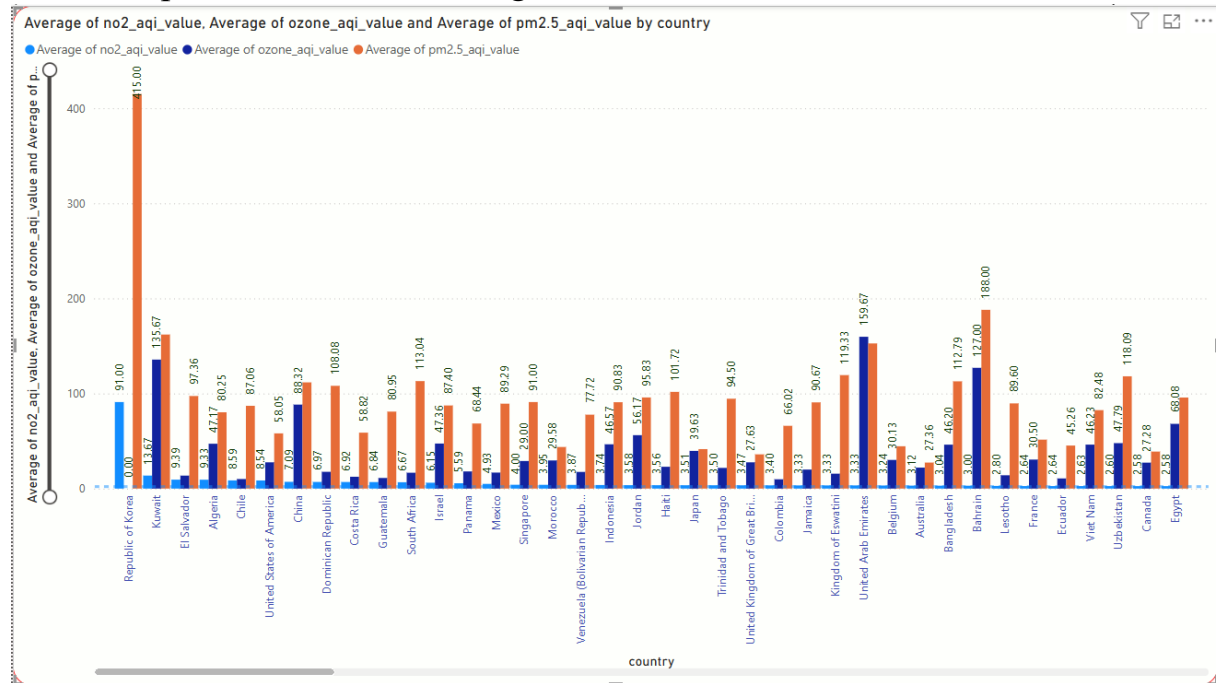
Displays average PM2.5, Ozone, and NO2 by country

Republic of Korea, Bahrain, and Pakistan rank among the most polluted



Column Chart

Compares NO2, Ozone, and PM2.5 averages country-wise
Shows Republic of Korea with highest PM2.5 at 415



Map Visual (Bubble Map)

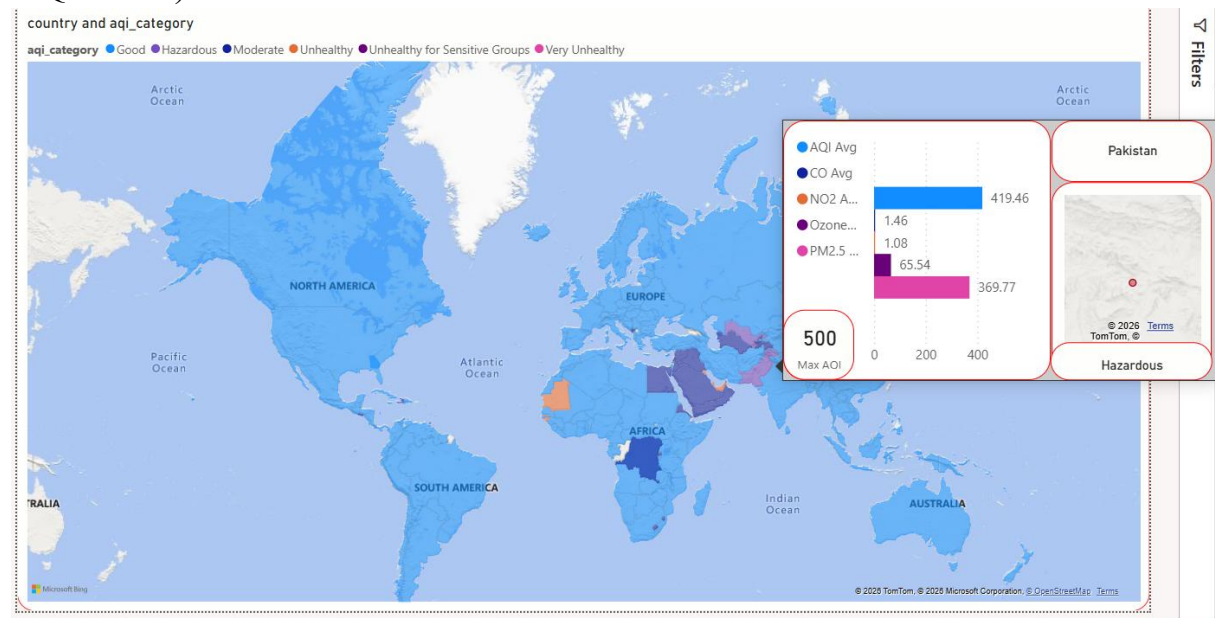
Shows average AQI value by country using bubble size
Tooltip displays AQI Avg, CO Avg, NO2 Avg, Ozone, PM2.5, and Max AQI per country



Choropleth Map

Displays country-wise AQI category using color coding

Middle East and South Asia regions show Hazardous levels (Pakistan Max AQI = 500)



Bar Chart

Average CO AQI value by country

Republic of Korea leads with 27.00

Q&A Visual

Allows natural language queries like "top countries by max CO" and "max AQI by country"

⚠ Q&A will be retiring in December 2026. Remove Q&A

Ask a question about your data

Try one of these to get started

what is the max AQI by country

what is the max co by country-city

what is the max co by air pollution data city

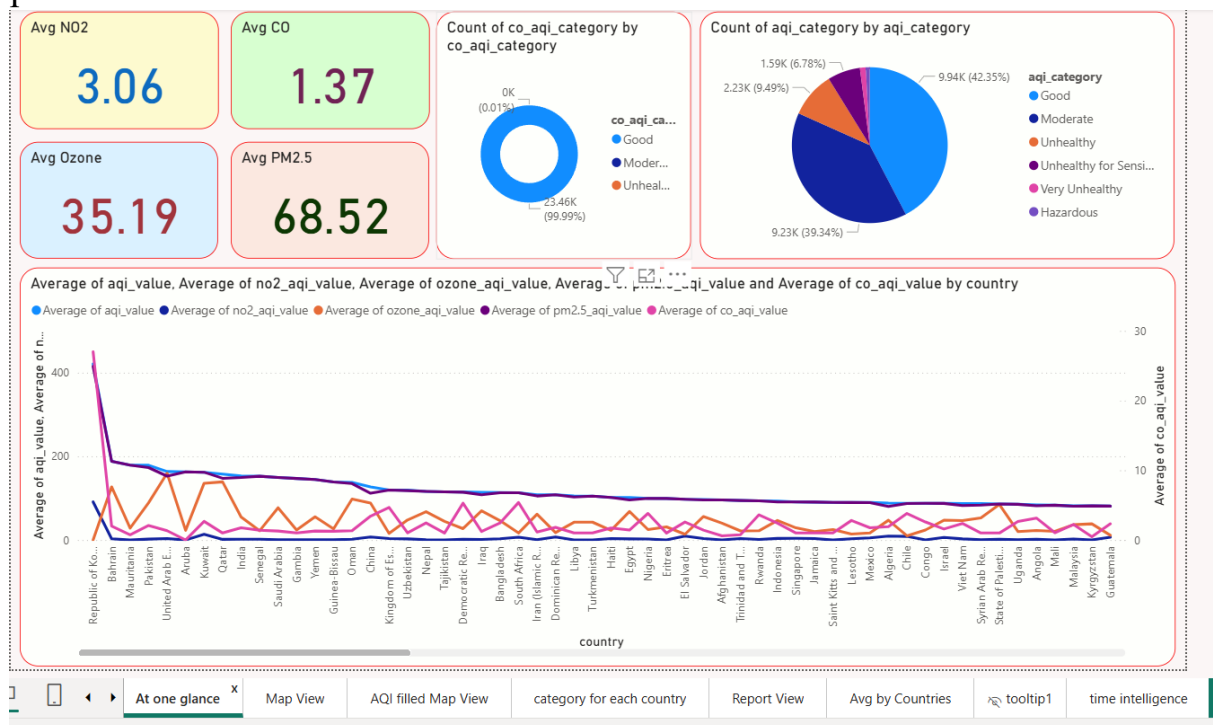
Show all suggestions

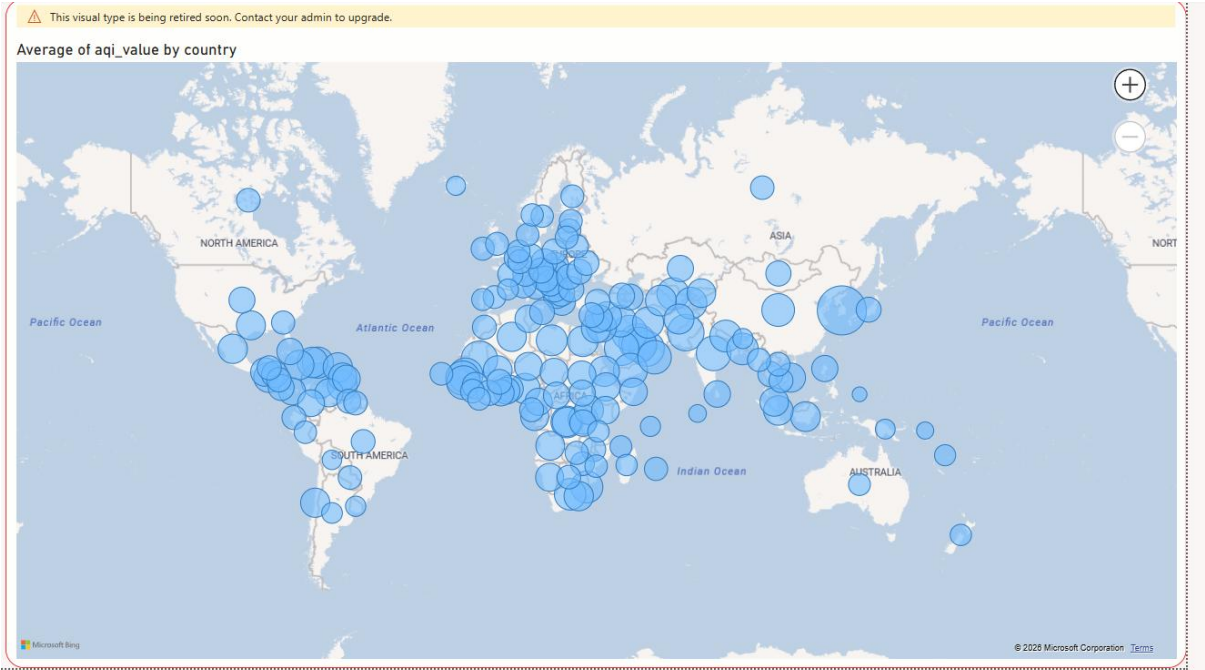
10. Interactivity Features

- Slicers for Country, AQI Category, CO Risk, AQI Risk Level
- Bookmarks for switching between dashboard pages
- Interactive visuals with click-based cross-filtering
- Tooltip cards showing country name, Max AOI, and AQI category on map hover

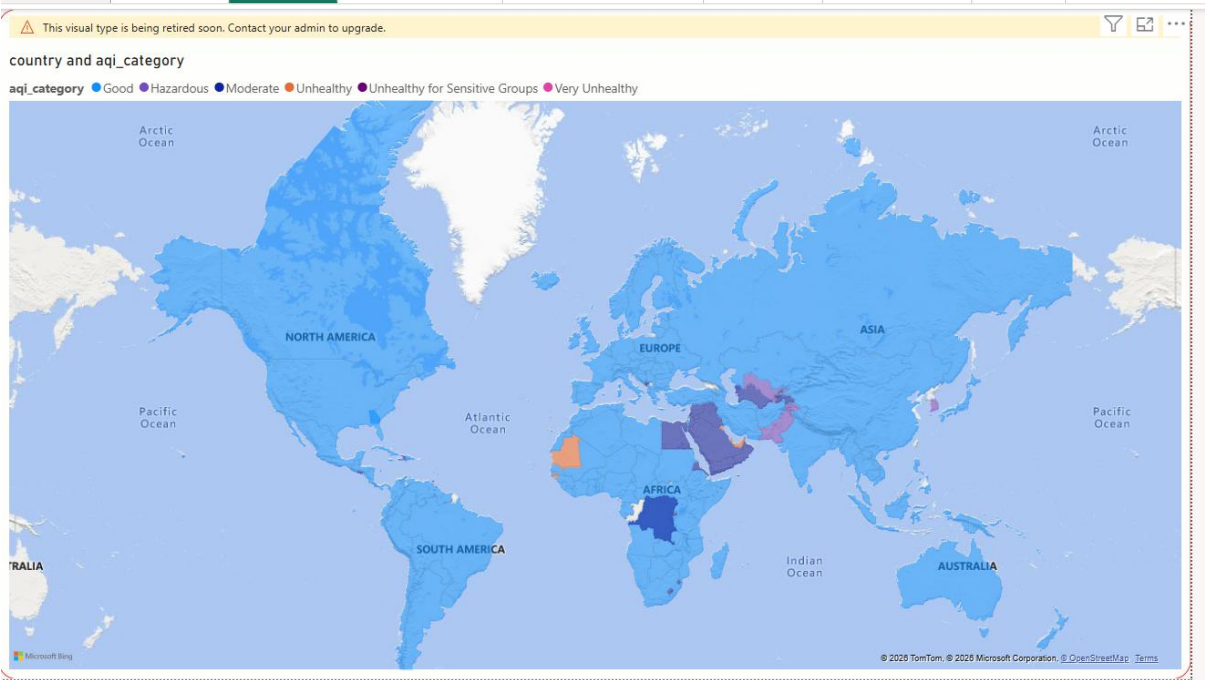
11. Map Visualization

The dataset contained actual city and country location data, which enabled real geographical map visuals. Both a bubble map (AQI by country) and a choropleth map (AQI category by country) were created to show global pollution distribution.

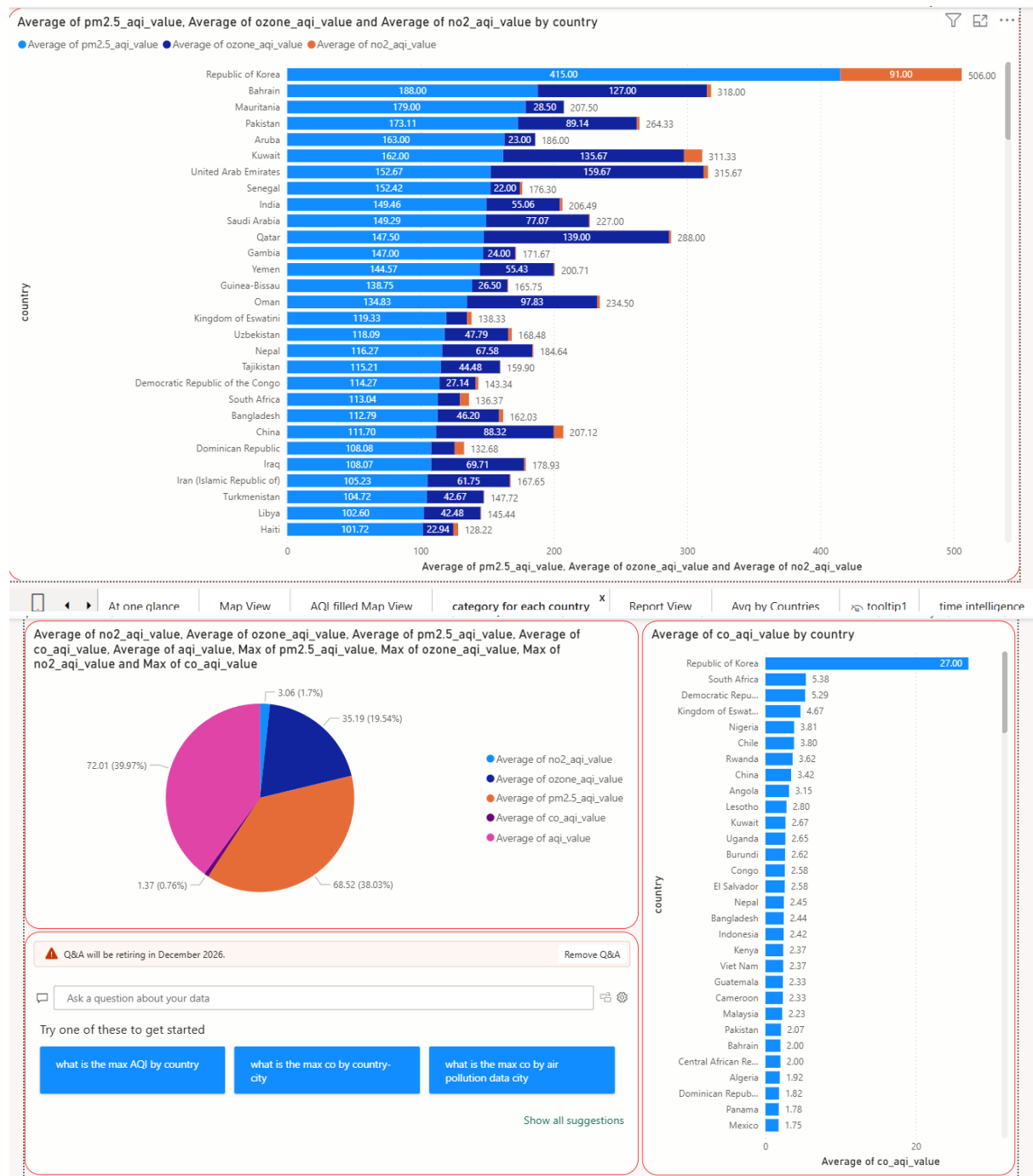


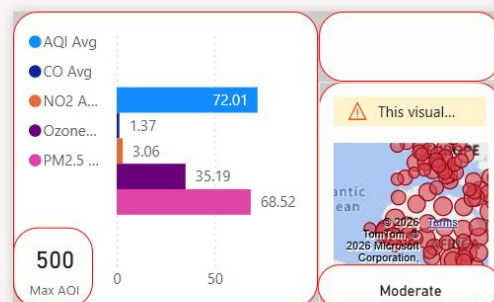
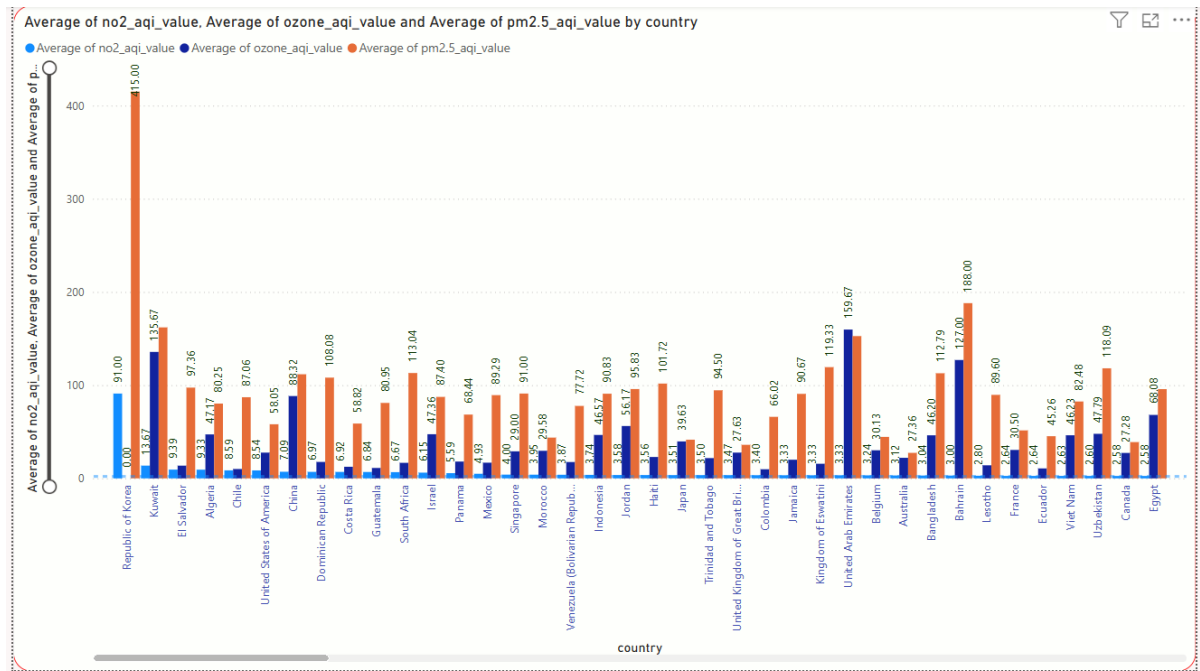


At one glance **Map View** x AQI filled Map View category for each country Report View Avg by Countries tooltip1 time intelligence



At one glance Map View **AQI filled Map View** x category for each country Report View Avg by Countries tooltip1 time intelligence





12. Key Insights

PM2.5 is the dominant pollutant globally, contributing ~38% of average AQI
 Republic of Korea, Bahrain, and Pakistan are among the most polluted countries

Pakistan recorded a maximum AQI of 500, classified as Hazardous

Majority of countries fall under the Good or Moderate AQI category (over 80%)

South Asia and the Middle East show consistently high pollution levels

CO levels are relatively low globally, with Republic of Korea being an outlier at 27.00 avg

13. SDG Goal Mapping

This project aligns with:

UN Sustainable Development Goal 3: Good Health and Well-being

UN Sustainable Development Goal 11: Sustainable Cities and Communities

UN Sustainable Development Goal 13: Climate Action

It helps in understanding global air pollution patterns and supports awareness for environmental decision-making.

14. Dashboard Design

- Clean multi-page layout with consistent color theme
- KPI cards placed at the top for quick summary
- Charts and maps arranged for logical flow of analysis
- Bookmarks used for easy navigation between pages

15. Conclusion

This project demonstrates the use of Power BI for global air pollution data analysis. By creating an interactive multi-page dashboard, meaningful insights were extracted from raw data regarding pollutant levels and country-wise AQI distribution.

The project highlights how data visualization can support better environmental decision-making and public health awareness.

16. Future Scope

Integrate real-time air quality API data

Apply machine learning models to predict AQI trends

Add time-series analysis with historical pollution data

Include city-level drill-down for more granular insights

Expand dashboard with health impact correlation analysis